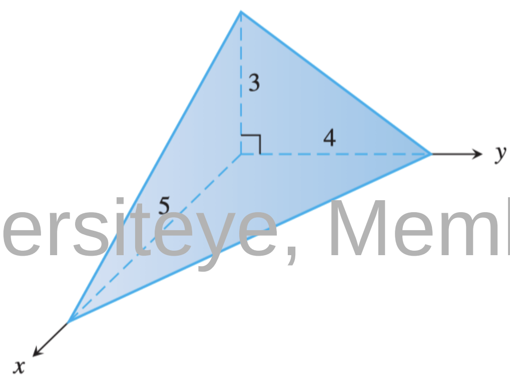
MAT 141 Basic Calculus I, Fall 2025-2026 Recitation #1

Q 1. *Find the volumes of the solids*

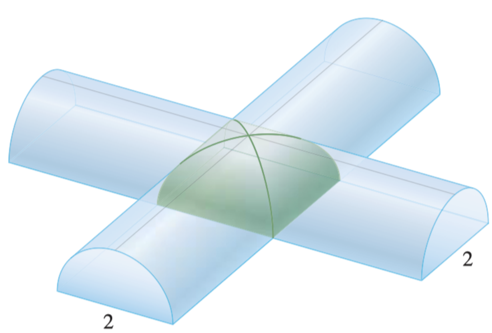
a) *The solid lies between planes perpendicular to the x-axis at $x = -1$ and $x = 1$. The cross-sections perpendicular to the x-axis are circular disks whose diameters run from the parabola $y = x^2$ to the parabola $y = 2 - x^2$.*

b) *The solid lies between planes perpendicular to the x-axis at $x = -1$ and $x = 1$. The cross-sections perpendicular to the x-axis between these planes are squares whose bases run from the semicircle $y = -\sqrt{1 - x^2}$ to the semicircle $y = \sqrt{1 - x^2}$.*

Q 2. *Find the volume of the given right tetrahedron*



Q 3. *Two half-cylinders of diameter 2 meet at a right angle in the accompanying figure. Find the volume of the solid region common to both half-cylinders.*



Q 4. *Find the volume of the solid generated by revolving the regions bounded by $y = \sqrt{9 - x^2}$, $y = 0$ about the $x-$axis*

Q 5. *Find the volumes of the solids enclosed by* $x = \sqrt{2\sin(2y)}$, $0 \leq y \leq \pi/2$, $x = 0$ *and revolved about the* $y-$*axis.*

Q 6. *Find the volumes of the solids generated by revolving the regions bounded by the lines and curves about the x-axis.*

a) $y = x^2 + 1$ *and* $y = x + 3$

b) $y = 4 - x^2$ *and* $y = 2 - x$

Q 7. *Find the volume of the solid generated by revolving the region bounded by* $y = 2\sqrt{x}$ *and the lines* $y = 2$ *and* $x = 0$ *about*

a. *the* x*-axis.*

b. *the* y*-axis.*

c. *the line* $y = 2$.

d. *the line* $x = 4$.